Sharp Poisson approximation and second-order correction for long multiplicative runs

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Abstract

Let f be completely multiplicative with independent fair signs at primes. We count starts of constant runs of length at least L in $[N, 2N)$, with $L = \log_2 N + O(1)$ and both signs included. The total variation distance to the Poisson law with nominal or exact mean has an explicit equivalent of order $LN^{-2/3}$, with a coefficient depending on $N2^{-L}$ and $L \bmod 3$. After retaining the exact mean and second factorial cumulant, the quadratic Poisson–Charlier approximation has optimal error power $4/3$: for every fixed $\varepsilon > 0$, its ℓ^1 error lies between constant multiples of $N^{-4/3-\varepsilon}$ and $N^{-4/3+\varepsilon}$, uniformly in each fixed critical band.

The leading correction comes from rational pairs. Keeping all channels up to a fixed height h gives an explicit distance equivalent with power $h/(h+1)$, determined by the first omitted height. The third factorial cumulant has optimal power $41/30$. The proof expands around the exact Bernoulli marginals and stops at the first excluded dependent configuration. Arithmetic bounds for fixed configurations and a separate uniform tail estimate control the resulting positive boundary mass. Short square relations and a common algebraic curve bound the triple interactions; a fourth Charlier test detects the two-pair contribution that makes the quadratic exponent optimal.

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1 Introduction

A long constant run in independent fair signs starts with probability 2^{-L}. Thus N possible starts have a bounded expected count when $L = \log_2 N + O(1)$. A completely multiplicative sign sequence has the same rare-event scale, but its distant windows share prime signs. The resulting dependencies leave an explicit signature in the Poisson approximation.

The first signature comes from pairs of windows in the ratio $2 : 3$. Their total contribution has size $LN^{-2/3}$. We determine its coefficient and then ask how accurately the whole counting law can be recovered after the exact pair correction is retained. The answer has optimal power $4/3$: two pair interactions survive at fourth order, whereas the total triple activity decays with power $41/30$. The gap $41/30 - 4/3 = 1/30$ is enough, provided that the expansion also controls configurations whose cardinality grows with N.

1.1 Model and principal results

Choose independent fair signs $f(\ell)$, indexed by primes, and set

$$f(n) = \prod_{\ell} f(\ell)^{v_\ell(n)} \qquad (n \geq 1).$$

The function f is completely multiplicative; in particular, $f(n^2) = 1$. Fix $C > 0$. Throughout, $N \geq 2$ and $L \geq 1$ are integers satisfying

$$I_N = [N, 2N) \cap \mathbb{Z}, \qquad |L - \log_2 N| \leq C, \qquad B = L + 1, \qquad p = 2^{-L}, \qquad \lambda_N = Np. \tag{1.1}$$

All logarithms without a subscript are natural. The nominal mean λ_N remains in a compact subinterval of $(0, \infty)$, but need not converge. Define

$$J_x = \mathbf{1}_{\{f(x-1)=-f(x),\ f(x)=f(x+1)=\cdots=f(x+L-1)\}}, \qquad Z_N = \sum_{x \in I_N} J_x. \tag{1.2}$$

Both signs of the run are included. The left boundary condition counts a long run once, at its start. A retained run may continue beyond $2N$, and a run starting before N is not counted.

Write

$$p_x = \mathbb{E}J_x, \qquad \mu_N = \mathbb{E}Z_N, \qquad \alpha_N = \mathrm{Var}(Z_N) - \mu_N.$$

The exact marginals p_x need not equal p. For a signed measure ν on $\mathbb{N} = \{0, 1, \ldots\}$, let $\|\nu\|_1 = \sum_{k \geq 0} |\nu\{k\}|$; total variation between probabilities is $d_{\mathrm{TV}}(P, Q) = \frac{1}{2}\|P - Q\|_1$. Set $\Pi_\mu = \mathrm{Pois}(\mu)$ and $\Delta = \delta_1 - \delta_0$. Products of measures are convolutions. For $r \geq 1$, write $(z)_r = z(z-1)\cdots(z-r+1)$, with $(z)_0 = 1$, and set $\Delta^{*0} = \delta_0$. The quadratic reference is

$$Q_{\mu,\alpha} = \Pi_\mu + \frac{\alpha}{2}\Pi_\mu * \Delta^{*2}. \tag{1.3}$$

It has total mass one and, in the range of the theorem below, is also nonnegative.

Theorem 1.1 (Optimal quadratic approximation with exact parameters)**.** *Fix $C > 0$. For every $\varepsilon > 0$, there are positive constants $c_{C,\varepsilon}$, $C_{C,\varepsilon}$ and a threshold $N_0(C, \varepsilon)$ such that, for all $N \geq N_0$ and every L satisfying (1.1),*

$$c_{C,\varepsilon} N^{-4/3-\varepsilon} \leq \|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \leq C_{C,\varepsilon} N^{-4/3+\varepsilon}. \tag{1.4}$$

Moreover, $\mu_N - \lambda_N = O_{C,\varepsilon}(N^{-1+\varepsilon})$, and Q_{μ_N,α_N} is a probability measure for all sufficiently large N, uniformly in the same band.

The theorem determines the optimal power of the remainder, with an arbitrary fixed power margin. It does not determine its logarithmic factor or leading constant. The exact mean matters: replacing it by λ_N can have a larger cost of order $N^{-1+o(1)}$. Proposition 7.4 makes this distinction precise.

The pair calculation gives a full leading equivalent. If $r \in \{0,1,2\}$ is the residue of L modulo 3, then

$$\alpha_N \sim \frac{r+2}{18\,2^{r/3}} \lambda_N^{5/3} L N^{-2/3}. \tag{1.5}$$

Here and below, an equivalent with moving parameters means that the ratio tends to one uniformly over the fixed critical band.

Corollary 1.2 (Exact Poisson total variation scale)**.** *For every $\varepsilon > 0$, uniformly under (1.1),*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \Pi_{\lambda_N}) = \frac{\alpha_N}{4} \|\Pi_{\lambda_N} * \Delta^{*2}\|_1 + O_{C,\varepsilon}(N^{-1+\varepsilon}). \tag{1.6}$$

Consequently,

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \Pi_{\lambda_N}) \sim \frac{r+2}{72\,2^{r/3}} \lambda_N^{5/3} \|\Pi_{\lambda_N} * \Delta^{*2}\|_1 \, L N^{-2/3}. \tag{1.7}$$

With Π_{μ_N} as the reference, the first formula holds with μ_N in the difference norm and error $O_{C,\varepsilon}(N^{-4/3+\varepsilon})$; the equivalent is unchanged.

Thus the power $2/3$, the factor L, and the leading coefficient are determined for both specified Poisson means. No minimization over all possible means is asserted.

Example 1.3 (A fixed phase). Take $N = 2^{3k}$ and $L = 3k$. Then $\lambda_N = 1$, $r = 0$, and $\|\Pi_1 * \Delta^{*2}\|_1 = 3/e$. Hence

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \Pi_1) \sim \frac{1}{12e} \, L N^{-2/3}. \tag{1.8}$$

The constant combines the arithmetic coefficient $1/9$, the second-difference norm $3/e$, and the factor $1/4$ from the cumulant and total variation conventions.

An explicit approximation can retain finitely many rational channels instead of the exact α_N. A channel $a : b$ has coprime integers $a < b < 2a$, and its height is b. Theorem 8.1 shows that keeping every height at most a fixed $h \geq 3$ gives a distance equivalent of order $LN^{-h/(h+1)}$, determined by the first omitted height. The complete $2:3$ and $3:4$ sums give the case $h = 4$, with optimal power $4/5$. In these statements h is fixed before $N \to \infty$. Theorem 3.3 also determines the optimal power $41/30$ of the third factorial cumulant; its positive witness consists of triples in the ratio $10 : 12 : 15$.

1.2 How the first channel produces its scale

The interval selects the first interior ratio. If $x \in I_N$, then $2x \notin I_N$, so the ratio $1:2$ has no macroscopic family of interior placements. The next ratio, $2:3$, does. For $L = 3m$, the windows at $2t$ and $3t$ contain the matched values $2(t+j)$ and $3(t+j)$, $0 \leq j < m$. For $j \neq k$,

$$f(2(t+j))f(2(t+k)) = f(3(t+j))f(3(t+k)).$$

The canonical matching has rank $m-1$ when there are no further square relations. Its prescriptions are compatible because the predecessor signs do not occur in these matches. Such a pair has

$$\mathbb{P}(J_{2t} = J_{3t} = 1) = 2^{-2L+m-1},$$

rather than the independent weight 2^{-2L}.

Allowing the offset $2y-3x$ to vary over a range of length $O(L)$ gives $O(NL)$ placements. Their typical joint weight is $N^{-5/3}$, which explains the product

$$(NL)N^{-5/3} = LN^{-2/3}.$$

This calculation identifies the mechanism. To obtain the coefficient, one must sum the exact endpoint and parity conditions and bound pairs with extra relations. Theorem 3.1 performs that calculation. The complete finite $2:3$ contribution matters: its next term, of order $N^{-2/3}$, is larger than the leading $3:4$ contribution of order $LN^{-3/4}$. Successive channels are therefore isolated by subtracting complete finite expressions.

1.3 From arithmetic interactions to the whole law

The reference is first a sum of independent Bernoulli variables with the exact probabilities p_x. This retains every one-window effect, including repeated indices. Each event is an affine condition on the prime signs over $\mathbb{F}_2$. Effective square-class spaces record its constraints after local relations have been removed. A homogeneous joint probability then dominates the exact rank weights, actual joint probabilities and absolute centered activities, so a single geometric estimate can control several terms in the expansion.

The finite stopping identity in Section 4 retains a hereditary class of configurations with a computable probability. It stops at the first excluded prefix and leaves the remaining indicators inside a positive conditional measure. The resulting error is controlled by the mass of that boundary. Fixed configurations are handled by conditional arithmetic counts. A separate certificate argument sums the large-cardinality tail. Bounded moments at every fixed order do not replace this uniform estimate.

For

$$A_3 = \sum_{\substack{T\subseteq I_N\\ |T|=3}} \left|\mathbb{E}\prod_{x\in T}(J_x - p_x)\right|,$$

Theorem 5.3 gives

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{\varepsilon}A_3 + N^{-4/3+\varepsilon}.$$

The arithmetic problem is therefore to place A_3 below the power $4/3$. Short square relations force a nonaligned pair onto a common algebraic curve. A degree-uniform integer-point estimate limits the rank shared by that pair. Combining this restriction with the explicit geometry of aligned triples gives $A_3 \ll N^{-41/30+\varepsilon}$. Taking a sufficiently small fixed margin yields

$$N^{\varepsilon}A_3 \ll N^{-4/3-(1/30-2\varepsilon)}, \qquad 0<\varepsilon<1/60.$$

This is the arithmetic gain that closes the upper bound.

For the lower bound, two pair interactions have size $\alpha_N^2 = N^{-4/3+o(1)}$. A fourth Charlier polynomial detects the corresponding term in the true law. The proof keeps a nonnegative compatible contribution and bounds the incompatible part before applying independently proved fixed-moment estimates. This establishes noncancellation for the specified quadratic reference. The exact Hermite reference already contains a fourth-order pair-pair term, so the same lower bound does not apply to it.

1.4 Context and organization

For independent signs, long runs and rare patterns have a classical Poisson theory; see Arratia, Gordon, and Waterman [2] and Godbole [10]. In the multiplicative model, Najnudel [14, Lemma 3.1 and Proposition 6.3] describes consecutive values through valuation relations and proves almost-sure frequencies of fixed words. Fixed-length frequencies do not supply the uniform rare-event

estimates needed when $L \asymp \log N$. The binary completely multiplicative model is also distinct from the squarefree-supported convention for a Rademacher multiplicative function; the former is called extended Rademacher in de la Bretèche, Wang, and Xu [7].

The preceding manuscript on this model establishes a Poisson approximation for the sign-resolved run field [16, Theorem 1.1] and already uses the $2:3$ interaction in a lower bound [16, Section 3.6, (3.22)]. Here the observable is the scalar count with both signs included. Its finer approximation requires the complete pair coefficient, control of the law below that coefficient, and a lower bound for the residual after correction.

Poisson approximation for dependent indicators has a general Chen–Stein framework [1]; asymptotic refinements include [3, 4]. The Poisson–Charlier organization by factorial cumulants is classical [5, Sections 5 and 8.1]. Čekanavičius and Vellaisamy [24, Theorems 3.1–3.2] distinguish quadratic and exponential corrections in a locally dependent setting. Their hypotheses do not cover the interactions between separated arithmetic windows here. The distance-equivalent theorem of Chhaibi, Delbaen, Méliot, and Nikeghbali [25, Theorem 3.11] uses a diverging parameter and regular Fourier residues; our Poisson means stay in a compact subset of $(0,\infty)$.

The finite stopping identity is a specialization of the abstract lace expansion [21]. Its application here depends on a retained class adapted to rational pairs and on arithmetic bounds for its boundary. Those bounds, the explicit channel coefficients and the noncancellation argument give the stated precision for multiplicative runs.

This article and its *Technical companion* form one work. References beginning with T belong to the companion. The article develops the exact finite expansion, states the arithmetic estimates it consumes, and carries out reconstruction and optimality. The companion proves the arithmetic and uniform counting estimates. Every error margin, fixed order and channel height is chosen before $N \to \infty$; any growing range has a separate uniform estimate.

2 Exact events and their relation spaces

The optimal remainder is smaller than N^{-1}, the cost of deleting all irregular windows. We therefore keep every event with nonzero probability from the outset. Its effective square-class space records all local relations once, and its actual probability is the correct centering parameter.

Let $\mathcal{I}_N = \{x \in I_N : p_x > 0\}$, where $p_x = \mathbb{E}J_x$. An event outside $\mathcal{I}_N$ is identically zero and can be removed without changing Z_N. For every $x \in I_N$, let S_x be the image of the even selections of $V_x = \{x-1,\dots,x+L-1\}$ in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, regarded as a vector space over $\mathbb{F}_2$. Write $d_x = \dim S_x$. The even-selection space has dimension L. Its kernel consists of local even square relations. The following bound makes the possible effective ranks explicit.

Lemma T1.18 in the companion proves that the local even square-relation code has dimension at most one in the critical band. Its proof combines a short-interval square-product bound with the binary Griesmer bound.

For an admissible window the affine prescription is consistent. Its probability is 2^{-d_x}, as the character expansion in Section 2.1 also shows. The lemma therefore gives

$$d_x \in \{L-1, L\}, \qquad p_x = 2^{-d_x} \in \{p, 2p\}, \qquad p_* := \max_{x\in\mathcal{I}_N} p_x \ll_C N^{-1}. \tag{2.1}$$

The assertion about d_x holds for all sufficiently large N in a fixed critical band. A local square relation may instead contradict the run prescription; that is precisely the case already removed by $p_x = 0$.

For a finite $I \subseteq \mathcal{I}_N$, put

$$\rho_{\mathrm{rel}}(I) = \sum_{x\in I} d_x - \dim \sum_{x\in I} S_x, \qquad d_{xy} = \dim(S_x \cap S_y), \tag{2.2}$$

$$q(I) = \mathbb{E} \prod_{x \in I} J_x, \qquad w(I) = 2^{\rho_{\mathrm{rel}}(I)} \prod_{x \in I} p_x, \tag{2.3}$$

$$c(I) = \mathbb{E} \prod_{x \in I} (J_x - p_x). \tag{2.4}$$

The empty products are one. We have $q(I) = w(I)$ when the affine prescriptions are jointly compatible, and $q(I) = 0$ otherwise. Write $G_N = \{x \in I_N : d_x = L\}$ for the regular set. On G_N, these definitions agree with the homogeneous-marginal notation used in the companion. Outside G_N, ρ_{rel} counts dependencies between effective spaces; it does not count local relations a second time.

The notions of directness, connectedness, and primitive family refer to the direct-sum decomposition of the effective spaces proved below. A family is direct when its effective spaces are in direct sum, and connected when it has just one canonical direct component. In particular, a primitive family has a nonzero relation and every proper subfamily is direct. Every nonzero relation of a primitive family has a nonzero projection in each of its blocks. A connected family need not have a relation with this property on its entire support. This distinction matters in the conditional count below.

For fixed $r \geq 2$, define

$$A_r = \sum_{\substack{I \subseteq \mathcal{I}_N \\ |I| = r}} |c(I)|, \qquad K_3^+ = \sum_{\substack{|T|=3,\ T \text{ connected} \\ q(T) > 0}} c(T). \tag{2.5}$$

The nonnegativity in the definition of K_3^+ will be proved in Proposition T9.3. The exact second and third factorial cumulants satisfy

$$\alpha_N = 2 \sum_{|e|=2} c(e) - \sum_x p_x^2, \tag{2.6}$$

$$\kappa_{3,N} = 6 \sum_{|T|=3} c(T) - 6 \sum_{e=\{x,y\}} (p_x + p_y) c(e) + 2 \sum_x p_x^3. \tag{2.7}$$

These identities follow by expanding the first three factorial moments. They display the repeated-index terms that an independent Bernoulli reference will retain exactly.

Fixed sources on effective spaces. The fixed-source estimates stated in Propositions T4.2 and T4.3 apply on $\mathcal{I}_N$, with the new conditional rank

$$\delta_z = \dim(S_z \cap \textstyle\sum_{x \in I} S_x)$$

and weight $p_z 2^{\delta_z}$. To see this, choose in each even-selection space a linear complement to its local kernel. The valuation map is injective on that complement and maps it onto S_x. Every label regularization, prime row, and disjoint square equation in the proofs is applied to this linear realization. Restriction to a complement restricts the allowed binary selections within the same physical coordinates; it does not enlarge any prime support. The factor p_z is at most $2p$, and the number of sources and targets is fixed in those propositions. Thus their constants change by a fixed factor. The same reasoning applies to homogeneous prescriptions. In particular, for fixed s, m and $\varepsilon > 0$,

$$\sum_{\substack{T \subseteq \mathcal{I}_N \setminus I \\ |T| = m}} q(I \cup T) \ll_{s,m,C,\varepsilon} N^{\varepsilon} q(I) \quad (|I| = s), \tag{2.8}$$

and the analogous homogeneous bound holds with q replaced by w. If $q(I) = 0$, every summand on the left is zero. No estimate here is asserted for s or m growing with N.

2.1 Affine structure and centered coefficients

The arithmetic estimates naturally count relations, whereas the approximation problem concerns a signed difference of laws. We introduce the exact identities linking these two objects. They will be used with a stopping rule, so that large connected families outside the required nullity range never need to be summed in absolute value.

Throughout this section $I \subseteq \mathcal{I}_N$, and $\rho_{\mathrm{rel}}(I)$ is the effective relative nullity. Write $b_x = \delta_0 + p_x\Delta$. The identities below hold for heterogeneous marginals. On the regular set G_N, one has $p_x = p$ and $d_x = L$. The same identities therefore cover both regular and irregular windows.

2.1.1 An exact expansion around independent Bernoulli variables

Set

$$Y_x = J_x - p_x, \qquad c(I) = \mathbb{E}\prod_{x\in I} Y_x, \qquad c(\varnothing) = 1.$$

Proposition 2.1 (Centered Bernoulli expansion)**.** *For every N,*

$$\mathcal{L}(Z_N) = \sum_{I\subseteq\mathcal{I}_N} c(I)\Delta^{*|I|} * \mathop{*}_{x\notin I} b_x. \tag{2.9}$$

The complementary convolution products are deterministic probabilities. The terms of sizes zero and two are

$$\mathop{*}_{x} b_x + \sum_{|e|=2} c(e)\Delta^{*2} * \mathop{*}_{x\notin e} b_x.$$

Terms of size one vanish.

Proof. For an indicator, $\delta_{J_x} = b_x + Y_x\Delta$. Expand the finite convolution product over $\mathcal{I}_N$ and take expectations. No independence assumption is used in this identity. □

Each $s \in S_x$ is a square-class vector. Let χ_s be its character on the prime signs, and let $\eta_x(s) \in \{\pm 1\}$ be the affine sign prescribed by the run-start event. The local prescription is consistent on $\mathcal{I}_N$, so its affine character η_x is well defined on the effective space. Thus

$$J_x = p_x \sum_{s\in S_x} \eta_x(s)\chi_s, \qquad \eta_x(0) = 1.$$

Lemma 2.2 (Full-support Fourier formula)**.** *For nonempty $I \subseteq \mathcal{I}_N$,*

$$c(I) = \Big(\prod_{x\in I} p_x\Big) \sum_{\substack{s_x\in S_x\setminus\{0\}\ (x\in I)\\ \sum_x s_x = 0}} \prod_{x\in I} \eta_x(s_x). \tag{2.10}$$

Consequently,

$$|c(I)| \le \Big(\prod_{x\in I} p_x\Big)(2^{\rho_{\mathrm{rel}}(I)} - 1) \le \Big(\prod_{x\in I} p_x\Big)2^{\rho_{\mathrm{rel}}(I)}.$$

If $c(I) \neq 0$, there is a relative relation meeting every block of I.

Proof. Subtracting p_x removes the zero character from each factor. The expectation of a character product is zero unless the sum of its indices vanishes. The kernel of $\bigoplus_{x\in I} S_x \to \sum_{x\in I} S_x$ has $2^{\rho_{\mathrm{rel}}(I)}$ elements. The displayed sum omits its zero element and possibly others. Every affine coefficient has modulus one. □

We extend the same formulas for $q(I)$, $c(I)$ and relative nullity to arbitrary original window families when stating the next lemma.

Lemma 2.3 (A common homogeneous envelope)**.** *For a nonempty family $I \subseteq I_N$ of original windows, set*

$$q_0(I) = 2^{-\dim(\sum_{x\in I} S_x)}, \qquad w(I) = \Big(\prod_{x\in I} p_x\Big) 2^{\rho_{\mathrm{rel}}(I)}.$$

Then

$$q_0(I) = 2^{\rho_{\mathrm{rel}}(I)} \prod_{x\in I} 2^{-d_x}, \qquad w(I),\ q(I),\ |c(I)| \le q_0(I). \tag{2.11}$$

These assertions allow locally inconsistent start prescriptions and zero marginals. In the critical band, for all sufficiently large N,

$$q_0(I) \le 2^{|I|} p^{|I|} 2^{\rho_{\mathrm{rel}}(I)}. \tag{2.12}$$

Proof. Rank–nullity gives the displayed factorization. A consistent affine prescription has the same probability as the corresponding homogeneous system; an inconsistent one has probability zero. Thus $p_x \le 2^{-d_x}$ and $q(I) \le q_0(I)$, proving the assertions for $w(I)$ and $q(I)$. If some $p_x = 0$, then $J_x = 0$ almost surely and $c(I) = 0$. Otherwise the full-support Fourier formula in Lemma 2.2 gives $|c(I)| \le w(I)$. Finally, local rigidity gives $d_x \ge L - 1$ uniformly on I_N in the critical band, so $2^{-d_x} \le 2p$ and the last inequality follows. □

The homogeneous quantity is an actual probability: it prescribes zero for every relative bit. It can be positive even when an original start is impossible. Its finite domination is valid at every cardinality; absorbing the factor $2^{|I|}$ into a constant requires a fixed cardinality.

2.1.2 The nullity that survives affine compatibility

Define

$$\rho_*(I) = \max\{\rho_{\mathrm{rel}}(A) : A \subseteq I,\ q(A) > 0\}, \qquad q(\varnothing) = 1, \quad \rho_{\mathrm{rel}}(\varnothing) = 0. \tag{2.13}$$

This parameter records the largest rank loss in a compatible subfamily. It can be smaller than the nullity of the whole family.

Lemma 2.4 (Compatible-nullity majorant)**.** *For every $I \subseteq \mathcal{I}_N$,*

$$|c(I)| \le 2^{|I|}\Big(\prod_{x\in I} p_x\Big) 2^{\rho_*(I)}, \qquad \rho_*(I) \le \rho_{\mathrm{rel}}(I). \tag{2.14}$$

Equality holds in the second inequality when $q(I) > 0$.

Proof. The exact inclusion-exclusion formula is

$$c(I) = \sum_{A\subseteq I} (-1)^{|I|-|A|} q(A) \prod_{x\in I\setminus A} p_x.$$

A compatible term has absolute value $(\prod_{x\in I} p_x) 2^{\rho_{\mathrm{rel}}(A)}$, and an incompatible term is zero. There are at most $2^{|I|}$ terms. Nullity increases under inclusion, because adding the block S_x increases the total rank by at most d_x. This proves the other assertions. □

2.1.3 Canonical direct components

A partition $\mathcal{P}$ of I is called direct if the spaces $V_C = \sum_{x\in C} S_x$, $C \in \mathcal{P}$, are in direct sum. Its finest such partition is the canonical direct decomposition of I. We call a nonempty family *connected* if this partition has only one part.

Lemma 2.5 (Existence, uniqueness, and factorization)**.** *Every finite family I has a unique finest direct partition $\mathcal{P}(I)$. For it,*

$$q(I) = \prod_{C\in\mathcal{P}(I)} q(C), \qquad c(I) = \prod_{C\in\mathcal{P}(I)} c(C), \qquad \rho_{\mathrm{rel}}(I) = \sum_{C\in\mathcal{P}(I)} \rho_{\mathrm{rel}}(C). \tag{2.15}$$

Proof. If $\mathcal{P}, \mathcal{Q}$ are direct partitions, their nonempty intersections form a direct partition. Indeed, inside each $P \in \mathcal{P}$, the spaces $\sum_{x\in P\cap Q} S_x$ lie in the mutually direct spaces V_Q, and sum to V_P. The direct sum over P proves the claim. Intersecting the finitely many direct partitions gives the unique finest one.

Characters indexed by mutually direct subspaces of the finite prime-sign vector space are independent uniform character families. Events, and centered variables, depending on different V_C are therefore independent. This proves both probability identities; rank additivity proves the last one. □

Connectedness here is defined by all relative relations. A minimal dependent family of original blocks is primitive, but primitive block supports alone do not determine the canonical components: an additional relation can join a third block to an already dependent pair. Exact pair compression will retain this possibility.

3 Pair and triple scales

The first deviation from Poisson is determined by rational coincidences between two separated windows. We keep the complete finite coefficient: its lower-order terms matter before the next channel becomes visible. The counting estimates behind this calculation are proved in Sections T1 and T2. For the pair profiles write $\rho(I) = |I|L - \dim\sum_{x\in I} S_x$ and let $\eta(I)$ indicate affine compatibility, so that $q(I) = p^{|I|}\eta(I)2^{\rho(I)}$. This raw nullity includes local kernels and is distinct from the effective relative nullity used in Section 2.

3.1 The two leading rational channels

An exact unit of the reduced channel $a : b$, $a < b$, is a pair of values (at, bt) in the two windows. A relative rational relation selects an even number of these units. If the first and last units are t and $t + d$, the number of selections is

$$A_1 = 1, \qquad A_d = 2^{d-2} \quad (d \geq 2).$$

Summing over positions in one window gives $L + 1 - D$ for a selection of diameter D. Retaining the predecessor sign changes this to

$$g_L(D) = (L - 1 - D)\mathbf{1}_{\{0\leq D\leq L\}}. \tag{3.1}$$

Indeed exactly one of the $L + 1 - D$ placements has its first selected vertex at the predecessor; its sign changes from $+1$ to -1. The value $g_L(L) = -1$ is part of the formula.

Define

$$U_{a,b}(L) = \sum_{d=1}^{\lfloor L/b\rfloor} A_d(L + 1 - ad)(L + 1 - bd),$$
$$S_{a,b}(L) = \sum_{d=1}^{\lfloor L/b\rfloor} A_d(L - 1 - ad)(L - 1 - bd).$$

For $L = bm + r$, $0 \le r < b$ and $m \ge 1$, both sums are evaluated by

$$\begin{aligned}\sum_{d=1}^{m} A_d(L+s-ad)(L+s-bd) \\ = 2^{m-1}\big[(b-a)(b+r+s)m + (r+s)^2 + (a+b)(r+s) + 3ab\big] \\ - (a+b)(L+s) - ab, \qquad s \in \{-1, 1\}.\end{aligned} \tag{3.2}$$

When $m = 0$ the defining sums are empty and equal to zero. To verify the identity for $m \ge 1$, expand the product and use

$$\sum A_d = 2^{m-1}, \quad \sum dA_d = (m-1)2^{m-1} + 1, \quad \sum d^2 A_d = (m^2 - 2m + 3)2^{m-1} - 1.$$

In particular, for $m \ge 1$,

$$S_{2,3}(L) = 2^{m-1}\big[(r+2)m + r^2 + 3r + 14\big] - 5L - 1, \qquad L = 3m + r, \quad 0 \le r < 3, \tag{3.3}$$

$$S_{3,4}(L) = 2^{m-1}\big[(r+3)m + r^2 + 5r + 30\big] - 7L - 5, \qquad L = 4m + r, \quad 0 \le r < 4. \tag{3.4}$$

Theorem 3.1 (Complete pair profile)**.** *The sums below are over ordered separated pairs in I_N:*

$$\sum_{|x-y|>L} (2^{\rho(x,y)} - 1) = \frac{N}{3}\{U_{2,3}(L) + U_{3,4}(L)\} + O_{C,\varepsilon}(N^{6/5+\varepsilon}), \tag{3.5}$$

$$\sum_{|x-y|>L} (\eta(x,y)2^{\rho(x,y)} - 1) = \frac{N}{3}\{S_{2,3}(L) + S_{3,4}(L)\} + O_{C,\varepsilon}(N^{6/5+\varepsilon}). \tag{3.6}$$

Writing $\mu_N = \mathbb{E}Z_N$ and $\alpha_N = \mathrm{Var}(Z_N) - \mu_N$, one has

$$\alpha_N = \frac{\lambda_N^2}{3N}\{S_{2,3}(L) + S_{3,4}(L)\} + O_{C,\varepsilon}(N^{-4/5+\varepsilon}). \tag{3.7}$$

Thus, with $r_3 = L \bmod 3$,

$$\alpha_N \sim \frac{r_3 + 2}{18\, 2^{r_3/3}} \lambda_N^{5/3} L N^{-2/3}. \tag{3.8}$$

After the entire first-channel term is removed, the next term is

$$\alpha_N - \frac{\lambda_N^2}{3N} S_{2,3}(L) \sim \frac{r_4 + 3}{24\, 2^{r_4/4}} \lambda_N^{7/4} L N^{-3/4}, \qquad r_4 = L \bmod 4. \tag{3.9}$$

The exact-unit calculation gives the displayed channel sums. Both orientations of each of the channels $2:3$ and $3:4$ have a translation interval of total length $N/3 + O(B)$. Private prime coordinates identify their rational kernels; the exceptional normalizations and all remaining channels have the stated residual mass. Section T2.4.3 gives the complete proof, including the endpoint, contact, and mean-transfer estimates.

The subtraction in (3.9) must retain the whole finite expression $S_{2,3}(L)$. Its own term of size $N^{-2/3}$ without the factor L is larger than $LN^{-3/4}$. The two exact channel polynomials together depend on $L \bmod 12$.

3.2 A computable two-channel reference

An entirely explicit reference has a different optimal rate. Retain the complete $2:3$ and $3:4$ channel sums, and set

$$\widehat{\alpha}_N = \frac{\lambda_N^2}{3N}\{S_{2,3}(L) + S_{3,4}(L)\}. \tag{3.10}$$

The sums have been defined above and are evaluated by (3.3)–(3.4).

Corollary 3.2 (An explicit probability reference)**.** *For every $\varepsilon > 0$, uniformly in the critical band,*

$$\left\| \mathcal{L}(Z_N) - \Pi_{\lambda_N} - \frac{\widehat{\alpha}_N}{2} \Pi_{\lambda_N} * \Delta^{*2} \right\|_1 \ll_{C,\varepsilon} N^{-4/5+\varepsilon}. \tag{3.11}$$

For all sufficiently large N, the reference $Q_{\lambda_N, \widehat{\alpha}_N}$ is a probability measure. Its mass at $k \geq 0$ is

$$e^{-\lambda_N} \frac{\lambda_N^k}{k!} \left[1 + \frac{\widehat{\alpha}_N}{2} \left\{ \frac{k(k-1)}{\lambda_N^2} - \frac{2k}{\lambda_N} + 1 \right\} \right]. \tag{3.12}$$

All parameters in this reference are determined by N, L.

Proof. Theorem 3.1 gives $\alpha_N - \widehat{\alpha}_N = O_{C,\varepsilon}(N^{-4/5+\varepsilon})$. Since $\|\Pi_{\lambda_N} * \Delta^{*2}\|_1 \leq 4$, replacing the coefficient costs at most $2|\alpha_N - \widehat{\alpha}_N|$. Apply Theorem 1.1. The replacement of μ_N by λ_N costs $O(N^{-1+\varepsilon})$, which is absorbed here. The finite formulas give $\widehat{\alpha}_N \sim \alpha_N > 0$ and $\widehat{\alpha}_N \to 0$, uniformly in the band. The positivity criterion in Lemma 7.1 therefore applies, since λ_N remains in a compact subinterval of $(0, \infty)$. □

The exponent $4/5$ in (3.11) is optimal for this two-channel reference. Its leading error is half the height-five contribution times $\|\Pi_{\lambda_N} * \Delta^{*2}\|_1$, as proved in Corollary 8.3. The explicit parameters do not supply a numerical value of the constant in the remainder or a numerical threshold for N. The same two-channel coefficient also gives a positive Hermite reference; see Corollary 8.4.

The probability of no retained start. Evaluating (3.11) at zero gives the immediate consequence

$$\mathbb{P}(Z_N = 0) = e^{-\lambda_N} \left(1 + \frac{\widehat{\alpha}_N}{2} \right) + O_{C,\varepsilon}(N^{-4/5+\varepsilon}), \tag{3.13}$$

because $(\Pi_{\lambda_N} * \Delta^{*2})\{0\} = e^{-\lambda_N}$. The positive correction shows an effect of the arithmetic pairs that is absent from the mean alone. Here $Z_N = 0$ means that no retained run start lies in I_N. A run starting before N may still intersect the interval, and a run starting in I_N may continue beyond $2N$. The exact-parameter refinement is Corollary 7.3.

3.3 Why triple interactions are smaller

The third factorial cumulant is

$$\kappa_{3,N} = \mathbb{E}(Z_N)_3 - 3\mu_N \mathbb{E}(Z_N)_2 + 2\mu_N^3. \tag{3.14}$$

It is distinct from the ordinary third cumulant, which equals the mean for a Poisson variable.

Theorem 3.3 (Optimal power of the third factorial cumulant)**.** *For each fixed critical band $C > 0$, there is a constant $c_C > 0$ such that, for all sufficiently large N,*

$$c_C N^{-41/30} \leq \kappa_{3,N} \ll_{C,\varepsilon} N^{-41/30+\varepsilon} \qquad (\varepsilon > 0). \tag{3.15}$$

Both bounds are unconditional. Thus $41/30$ is the optimal power exponent for this factorial cumulant. The theorem does not assert an asymptotic equivalent or identify a logarithmic factor for the full cumulant.

The complete proof is given in Section T10. It bounds all admissible triples, with the exact marginals retained. The following form is the input to the reconstruction.

Corollary 3.4 (Absolute third-order activity)**.** *For every $\varepsilon > 0$,*

$$A_3 := \sum_{|T|=3} |c(T)| \ll_{C,\varepsilon} N^{-41/30+\varepsilon}. \tag{3.16}$$

Proof. Equation (T9.1) gives $A_3 = K_3^+ + O(N^{-3/2+\varepsilon'})$. Use the upper bound of Theorem 3.3 and reduce the internal margin. □

The configuration $10t, 12t, 15t$ explains the third-order power. Multiplication by $6, 5, 4$ gives a common center $60t$ and offset steps $6, 5, 4$. The canonical relative nullity is

$$\left\lfloor \frac{L-1}{6} \right\rfloor + \left\lfloor \frac{L-1}{3} \right\rfloor + \left\lfloor \frac{L-1}{5} \right\rfloor - \left\lfloor \frac{L-1}{15} \right\rfloor = \frac{19}{30}L + O(1).$$

A compatible triple with no additional square relations therefore has weight $N^{-71/30+o(1)}$. There are $N/30 + O(1)$ translations in the dyadic interval. Proposition T10.4 checks the private coordinates, the predecessor signs and the positivity of the centered coefficient; it makes the lower bound from this family rigorous. The upper bound also needs pair rigidity and an exhaustive partition of the other triples. It is not a consequence of this example alone.

The distinction between pairs and triples is decisive: $\alpha_N^2 = N^{-4/3+o(1)}$, whereas $A_3 \ll N^{-41/30+\varepsilon}$ and $41/30 - 4/3 = 1/30$. The reconstruction will retain the first pair term and allow its square in the error. The fourth-order test will prove that this remaining scale cannot be improved in power for the quadratic reference.

4 Stopping the expansion at its first excluded prefix

The leading correction comes from retaining one dependent pair. An expansion of the entire counting law would also retain every possible interaction after that pair. Taking absolute values at that stage can be expensive, even when the interactions that first leave the retained class have small total probability. We therefore stop each term at its first excluded prefix and leave all subsequent indicators inside a probability kernel.

The first identity is finite and does not require independence. Its exact-marginal application will reduce the approximation problem to families whose effective nullity is at most $2L$. The first-excluded-prefix identity is a specialization of Zeilberger's abstract lace expansion [21, formula (*)]. The broader inclusion–exclusion context includes the abstract-tube identities [13, 8]. We give the prefix proof and its convolution form; the model-specific estimates concern the retained arithmetic class and its full boundary.

4.1 An identity for an arbitrary hereditary class

Let G be a finite totally ordered set. A family $\mathscr{D} \subseteq 2^G$ is hereditary if $I \in \mathscr{D}$ and $J \subseteq I$ imply $J \in \mathscr{D}$. Assume throughout that $\varnothing \in \mathscr{D}$. Its *ordered boundary* is

$$\partial\mathscr{D} = \{I \subseteq G : I \notin \mathscr{D},\ I \setminus \{\max I\} \in \mathscr{D}\}. \tag{4.1}$$

This definition records the first excluded prefix in the prescribed order. It is not the family of all inclusion-minimal excluded sets: an element of $\partial\mathscr{D}$ can contain a smaller excluded subset which is not a prefix.

Lemma 4.1 (Expansion at the first excluded prefix)**.** *Let $(u_x)_{x\in G}$ belong to a commutative unital algebra. Then, as a polynomial identity,*

$$\prod_{x\in G}(1+tu_x) = \sum_{I\in\mathscr{D}} t^{|I|}\prod_{x\in I}u_x + \sum_{I\in\partial\mathscr{D}} t^{|I|}\prod_{x\in I}u_x \prod_{\substack{y\in G\\ y>\max I}}(1+tu_y). \tag{4.2}$$

Proof. Expand the product on the left and index its monomials by $S \subseteq G$. If $S \notin \mathscr{D}$, read the elements of S in their prescribed order. There is a unique first prefix $I \notin \mathscr{D}$. Its preceding prefix belongs to $\mathscr{D}$, so $I \in \partial\mathscr{D}$, and every element of $S \setminus I$ is larger than $\max I$.

Conversely, take $I \in \partial\mathscr{D}$ and $T \subseteq \{y \in G : y > \max I\}$. Every proper prefix of I belongs to $\mathscr{D}$, by heredity. The set $I \cup T$ is excluded, since it contains I, and its first excluded prefix is I. Thus the pairs (I, T) partition the monomials indexed by $S \notin \mathscr{D}$. Summing these monomials proves the identity. □

To identify the specialization explicitly, define $\ell(S) = S$ for $S \in \mathscr{D}$, and otherwise let $\ell(S)$ be its first excluded prefix. This is a lace map: $\ell(S) \subseteq S$; $\ell(T) = \ell(S)$ whenever $\ell(S) \subseteq T \subseteq S$; and equal images of two sets remain unchanged under their union. Its fixed points are $\mathscr{D} \cup \partial\mathscr{D}$. The compatible-extension set is empty for $I \in \mathscr{D}$, and equals $\{y \in G : y > \max I\}$ for $I \in \partial\mathscr{D}$. Zeilberger's formula (*) with $Y_x = tu_x$ is therefore exactly (4.2).

For indicators $(J_x)_{x\in G}$, write

$$Z = \sum_{x\in G} J_x, \qquad q(I) = \mathbb{E}\prod_{x\in I} J_x, \qquad q(\varnothing) = 1.$$

Let $\Delta = \delta_1 - \delta_0$, and use the convention $\Delta^{*0} = \delta_0$. The measure retained by $\mathscr{D}$ is

$$T_{\mathscr{D}} = \sum_{I\in\mathscr{D}} q(I)\Delta^{*|I|}. \tag{4.3}$$

It has total mass one, although its individual masses need not be nonnegative.

Proposition 4.2 (A positive remainder kernel)**.** *For arbitrary dependence among the indicators,*

$$\mathcal{L}Z - T_{\mathscr{D}} = \sum_{I\in\partial\mathscr{D}} \Delta^{*|I|} * \nu_I, \tag{4.4}$$

$$\nu_I = \mathbb{E}\left[\left(\prod_{x\in I} J_x\right)\delta_{\sum_{y>\max I} J_y}\right].$$

The measure ν_I is positive and has mass $q(I)$. Consequently,

$$\|\mathcal{L}Z - T_{\mathscr{D}}\|_1 \le \sum_{I\in\partial\mathscr{D}} 2^{|I|} q(I). \tag{4.5}$$

Proof. Apply Lemma 4.1 in the convolution algebra of finite signed measures, with unit δ_0 and $t = \Delta$. For an indicator,

$$\delta_0 + J_x\Delta = \delta_{J_x}, \qquad \underset{y>\max I}{*}(\delta_0 + J_y\Delta) = \delta_{\sum_{y>\max I} J_y}.$$

The expectation of the complete product is $\mathcal{L}Z$. Taking expectations in (4.2) therefore gives (4.4). Positivity and the mass formula for ν_I follow directly from its definition. Finally, $\|\Delta^{*|I|}\|_1 = 2^{|I|}$ and the convolution norm is submultiplicative, which proves (4.5). □

If $q(I) > 0$, the normalized measure $\nu_I/q(I)$ is the law of the suffix count conditioned on the activity of every member of I. This conditional law can differ substantially from its unconditional version. The proof uses only its total mass, so no independence between the prefix and suffix is required. In particular, it incurs no factor exponential in the number of suffix indicators. The use of the ordinary ℓ^1 norm matters: an exponential weight would also charge the suffix count.

Example 4.3 (A suffix that remains fully dependent). Take $G = \{1, 2, 3\}$ in its natural order, and retain only the empty set and the singletons. Then

$$\partial\mathscr{D} = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}, \qquad T_{\mathscr{D}} = \delta_0 + (q_1 + q_2 + q_3)\Delta,$$

and the exact remainder is

$$\mathcal{L}Z - T_{\mathscr{D}} = \Delta^{*2} * \left(\mathbb{E}[J_1J_2\delta_{J_3}] + (q_{13} + q_{23})\delta_0\right).$$

For $J_1 = J_2 = J_3 = H$, where $\mathbb{P}(H = 1) = \theta$, the conditional suffix following $\{1, 2\}$ is identically one. The formula becomes

$$\mathcal{L}(3H) - (\delta_0 + 3\theta\Delta) = \theta\Delta^{*2} * (2\delta_0 + \delta_1).$$

The remainder has norm 6θ, while (4.5) gives the valid bound 12θ. The mechanism is the positive kernel of mass 3θ, not a comparison of the conditional suffix with an independent one. This finite example explains the stopping identity; the remaining task in our model is to prove that the appropriate boundary has small weighted mass.

5 Reconstructing the law with every marginal retained

At a precision finer than N^{-1}, removing the exceptional windows is too expensive. We retain their exact marginals instead. This also keeps the repeated occurrence of a single Bernoulli variable inside the reference law, where it can be evaluated exactly. The remaining task is to control interactions between distinct windows and to sum the expansion without applying a fixed-order estimate at an order growing with N.

We use the full admissible set $\mathcal{I}_N = \{x \in I_N : p_x > 0\}$, the exact marginals $p_x = \mathbb{E}J_x$, the centered coefficients $c(I) = \mathbb{E}\prod_{x\in I}(J_x - p_x)$, and the relative nullity defined in Section 2. Throughout this section we abbreviate $\rho_{\mathrm{rel}}(I)$ to $\rho(I)$; this convention uses the effective local ranks and does not add the local rank defects again. Put

$$A_2 = \sum_{|e|=2} |c(e)|, \qquad A_3 = \sum_{\substack{|T|=3 \\ T \text{ connected}}} |c(T)|, \qquad p_* = \max_{x\in\mathcal{I}_N} p_x.$$

The marginal and pair estimates give $p_* \le 2p \ll_C N^{-1}$, $\mu = \sum_x p_x = O_C(1)$, and $A_2 \ll_{C,\varepsilon} N^{-2/3+\varepsilon}$. The connection between the arithmetic estimates and the distribution is the following bound:

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{\varepsilon}A_3 + N^{-4/3+\varepsilon}. \tag{5.1}$$

We prove it in Theorem 5.3. The two terms have different origins: the first records a genuinely connected three-window interaction, while the second permits two pair interactions.

Three errors in the reconstruction. For $e = \{x, y\}$ write $S_e = S_x + S_y$. The retained class $\mathscr{D}_2$ consists of direct families and families $e \cup U$ with a dependent pair e for which S_e and the spaces $(S_z)_{z\in U}$ form a direct sum. The dependent pair is then unique. Write $T_{\mathscr{D}_2} = \sum_{I\in\mathscr{D}_2} q(I)\Delta^{*|I|}$. With the covariance-corrected Bernoulli measure B_2 defined in (5.2), the proof separates

$$\begin{aligned}\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \le \|\mathcal{L}(Z_N) - T_{\mathscr{D}_2}\|_1 + \|T_{\mathscr{D}_2} - B_2\|_1 \\ + \|B_2 - Q_{\mu_N,\alpha_N}\|_1.\end{aligned}$$

The first term is the stopping boundary, with the unexpanded suffix kept in a positive conditional measure. The second consists of the independent and pair deficits in (5.8); the covariance signs are retained before taking norms. In particular, an incompatible pair is allowed in $\mathscr{D}_2$: its joint probability is zero, but its covariance is still present in B_2. The last term costs $O(N^{-5/3+\varepsilon})$ by Lemma 5.1; that rate concerns the full measure B_2, including every covariance. Theorem 5.2 assembles the three estimates, using fixed orders and the separately uniform tail of Proposition T12.11 for the boundary.

5.1 The exact Bernoulli reference and its conversion

For any finite family of Bernoulli variables, without an independence assumption, define $b_i = \delta_0 + p_i\Delta$ and

$$B_2 = \mathop{*}_{i\in\mathcal{I}_N} b_i + \sum_{|e|=2} c(e)\Delta^{*2} * \mathop{*}_{i\notin e} b_i. \tag{5.2}$$

This is a signed measure. Its first term is the law of a sum of independent Bernoulli variables with the exact individual marginals; its second term restores every covariance once.

Lemma 5.1 (Conversion with heterogeneous marginals)**.** *Let* $s_j = \sum_i p_i^j$, $C_2 = \sum_{|e|=2} c(e)$, *and* $\alpha = 2C_2 - s_2$. *If* $\mu \le M$ *and* $p_* \le 1/4$, *then*

$$\|B_2 - Q_{\mu,\alpha}\|_1 \le C_M\{s_3 + s_2^2 + A_2(s_2 + p_*)\}. \tag{5.3}$$

For the present model this is $O_{C,\varepsilon}(N^{-5/3+\varepsilon})$.

Proof. In the convolution algebra of finite signed measures, the series for $\log_*(\delta_0 + p_i\Delta)$ converges absolutely, since $\|p_i\Delta\|_1 = 2p_i \le 1/2$. Summing it gives

$$\log_*\Big(\mathop{*}_i b_i\Big) = \mu\Delta - \frac{s_2}{2}\Delta^{*2} + E, \qquad \|E\|_1 \le \sum_{j\ge 3}\frac{2^j s_j}{j} \le \frac{16}{3}s_3.$$

Exponentiation and $\exp_*(\mu\Delta) = \Pi_\mu$ imply

$$\Big\|\mathop{*}_i b_i - \Pi_\mu + \frac{s_2}{2}\Pi_\mu * \Delta^{*2}\Big\|_1 \le C_M(s_3 + s_2^2).$$

The product omitting a pair $e = \{i, j\}$ differs from $\Pi_{\mu - p_i - p_j}$ by $O_M(s_2)$ in norm, and

$$\|\Pi_{\mu-p_i-p_j} - \Pi_\mu\|_1 \le 2(p_i + p_j).$$

Replacing each such product in (5.2) by Π_μ therefore costs $O_M(A_2(s_2 + p_*))$. The exact identity $\operatorname{Var} Z - \mathbb{E}Z = 2C_2 - s_2$ proves the lemma. Here $s_2 \le p_*\mu = O_C(N^{-1})$ and $s_3 \le p_*^2\mu = O_C(N^{-2})$, giving the stated application. □

Algebraic form of the conversion. The estimate in Lemma 5.1 also holds for arbitrary real pair weights $c(e)$ in (5.2), with $C_2 = \sum_e c(e)$, $A_2 = \sum_e |c(e)|$ and $\alpha = 2C_2 - s_2$. The mean and marginal bounds are unchanged. The two approximations in its proof and the triangle inequality give this algebraic form; only the interpretation $\alpha = \operatorname{Var} Z - \mathbb{E}Z$ uses actual covariances.

5.2 Arithmetic estimates used in the assembly

The proof below uses three estimates proved in Section T12 of the companion. They separate fixed geometry from the growing number of windows. For each fixed $r \ge 4$, the total connected activity is

$$\sum_{\substack{|I|=r\\ I \text{ connected}}} |c(I)| \ll_{r,C,\varepsilon} N^{\varepsilon}A_3 + N^{-4/3+\varepsilon},$$

and its incompatible part is $O(N^{-3/2+\varepsilon})$. The independent deficit of all pair extensions is $O(N^{-3/2+\varepsilon})$ at every cardinality. Finally, the connected activity beyond the fixed size 256, up to the logarithmic boundary cutoff and with compatible nullity at most $2L$, is $O(N^{-4/3+\varepsilon})$; the corresponding primitive tail has the same bound without a cardinality cutoff. These are Proposition T12.9, Proposition T12.8 and Proposition T12.11, respectively.

The last estimate has a separate proof. It uses small-kernel counts, moderate congruence certificates and a graph with small incoming and outgoing degrees. It does not let the order in the first estimate grow with N. The choice 256 is sufficient for these estimates; no optimality is attached to that cutoff.

5.3 The stopped expansion and its finite input

Let $\mathscr{D}_2$ consist of the direct families and the families $e \cup U$ for which e is any dependent pair and $S_e, (S_z)_{z\in U}$ are direct. Affine incompatibility of e is allowed. This class is hereditary; every non-direct member contains exactly one dependent pair. For a fixed ordering of $\mathcal{I}_N$, its boundary contains the first excluded prefixes. A boundary family has relative nullity at most $2L$: its retained predecessor has at most L, and adding one block adds at most L. In particular every subfamily of a boundary family has compatible nullity at most $2L$.

Theorem 5.2 (Finite transfer to the complete law)**.** *Fix $1 < \beta \le 4/3$. Suppose that, for each of the fixed integers $3 \le r \le 256$ and every $\varepsilon > 0$,*

$$\sum_{\substack{|I|=r,\ I \text{ connected} \\ \rho_*(I)\le 2L}} |c(I)| \ll_{r,C,\varepsilon} N^{-\beta+\varepsilon}. \tag{5.4}$$

Then

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{-\beta+\varepsilon}. \tag{5.5}$$

Only these 254 fixed orders are assumed in (5.4); the larger orders are supplied by Proposition T12.11.

Proof. Fix a weight $w > 0$. The hypothesis at sizes at most 256 and (T12.17) give, for $K_N = o(L)$,

$$T := \sum_{\substack{I \text{ connected},\ 3\le|I|\le K_N \\ \rho_*(I)\le 2L}} w^{|I|}|c(I)| \ll N^{-\beta+\eta}. \tag{5.6}$$

The finitely many factors w^r at $r \le 256$ only change constants. For a primitive P, $\rho(P) \le L$ and $\prod_{x\in P} p_x \le |c(P)|$, because $c(P) = q(P) - \prod_P p_x$ and $q(P)$ is either zero or at least twice the independent product. Together with (T12.18), this gives

$$\sum_{\substack{P \text{ primitive} \\ |P|\ge 3}} w^{|P|}|c(P)| \ll N^{-\beta+\eta} \tag{5.7}$$

over all cardinalities.

Put $T_{\mathscr{D}_2} = \sum_{I\in\mathscr{D}_2} q(I)\Delta^{*|I|}$. The finite expansion of B_2 yields the identity

$$T_{\mathscr{D}_2} - B_2 = -\sum_{I\notin\mathscr{D}_2} \Delta^{*|I|}\left[\prod_{i\in I} p_i + \sum_{\substack{e\subset I \\ |e|=2}} c(e) \prod_{i\in I\setminus e} p_i\right]. \tag{5.8}$$

For the independent term, choose a primitive subfamily of a discarded family. If it has at least three members, use (5.7); if it is a pair e, the complement must be a non-direct extension of S_e, and $\prod_{i\in e} p_i \le |c(e)|$. The pair terms in (5.8) are bounded by the same all-cardinality pair deficit. Free complements have total weight at most $e^{2\mu}$. Consequently

$$\|T_{\mathscr{D}_2} - B_2\|_1 \ll N^{-\beta+\eta}. \tag{5.9}$$

Take $K_N = (5 + o(1)) \log N/\log\log N$. The exact stopping identity from Section 4 gives

$$\|\mathcal{L}(Z_N) - T_{\mathscr{D}_2}\|_1 \le \sum_{I\in\partial\mathscr{D}_2} 2^{|I|} q(I).$$

For $|I| \le K_N$, expand $q(I)$ in centered coefficients. The terms of size zero and two are the independent and pair deficits already bounded. Every remaining nonzero centered term has

either a connected component of size at least three or at least two pair components. Canonical factorization, (5.6), and the free independent complements bound their sum by a constant times

$$e^{4A_2+T} - 1 - 4A_2 = O(T + A_2^2).$$

Here $T = o(1)$ and $A_2^2 \ll N^{-4/3+\eta}$; the centered coefficient of a singleton is zero. This is where the linear contribution of one pair is subtracted before taking an absolute bound.

On the larger boundary, $q(I) \le 2^{2L} \prod_I p_x$, so

$$\sum_{\substack{I \in \partial \mathscr{D}_2 \\ |I| > K_N}} 2^{|I|} q(I) \ll_C N^2 \sum_{r > K_N} \frac{(2\mu)^r}{r!} = N^{-3+o(1)}.$$

Combining these bounds, (5.9), and Lemma 5.1 proves (5.5). All internal margins are chosen after the final ε; none is made to vary with N. □

Theorem 5.3 (Reconstruction from triple activity)**.** *With the exact mean and second factorial cumulant,*

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{\varepsilon} A_3 + N^{-4/3+\varepsilon}. \tag{5.10}$$

In particular the arithmetic triple estimate gives the upper bound $O_{C,\varepsilon}(N^{-4/3+\varepsilon})$ for the full model.

Proof. In the proof of Theorem 5.2, replace the finite activity input by Proposition T12.9. At sizes $3, \dots, 256$ its total cost is $N^{\varepsilon'} A_3 + N^{-4/3+\varepsilon'}$; the larger sizes have the second cost by Proposition T12.11. The primitive deficits, pair deficits, and boundary expansion obey the same bounds. In the component generator,

$$T \ll N^{\varepsilon'} A_3 + N^{-4/3+\varepsilon'}, \qquad e^{4A_2+T} - 1 - 4A_2 = O(T + A_2^2).$$

The established arithmetic estimate makes $T = o(1)$, so this passage loses no new fixed power of N. The large boundary costs $N^{-3+o(1)}$ and conversion costs $N^{-5/3+\varepsilon'}$. To make the choices explicit, denote the requested final margin by $\varepsilon_{\mathrm{out}} = \varepsilon$ and choose $0 < \tau < \min(\varepsilon_{\mathrm{out}}/4, 1/960)$. Invoke each completed fixed-order estimate with sufficiently small fixed input margins that its total displayed loss is at most τ. For example, use the pair estimate with loss $\tau/2$ before squaring A_2. Only the fixed orders $3, \dots, 256$ occur in this choice; the larger orders use the uniform certificate bounds, not constants from a fixed-order theorem. With $\varepsilon' = \tau$ the preceding argument proves (5.10), since $\tau < \varepsilon_{\mathrm{out}}$.

Finally invoke Corollary 3.4 with loss τ. The triple term is then at most $N^{-41/30+2\tau}$, and

$$\frac{41}{30} - 2\tau - \frac{4}{3} = \frac{1}{30} - 2\tau > 0.$$

The conversion and large-boundary exponents exceed $4/3$ by $1/3$ and $5/3$, respectively. These comparisons give the announced upper bound. All accuracy choices, source counts and fixed activity orders used here are fixed before N; the growing cardinality cutoff K_N is controlled separately. □

Power margins. The strict power savings used in the last stages are recorded below. Each margin compares complete contributions, after coefficient choices, propagation and rank factors have been included.

Comparison	Exact margin	Location
Nonaligned target above a rational pair versus dominant triples	$11/8 - 41/30 = 1/120$	(T10.11)
Incompatible triples versus the third factorial cumulant	$3/2 - 41/30 = 2/15$	(T9.1)
Third activity versus the quadratic remainder	$41/30 - 4/3 = 1/30$	(5.10)
Low-nullity uniform tail versus the quadratic remainder	$27/20 - 4/3 = 1/60$	(T12.20)
Quartic signed error versus α_N^2	$3/2 - 4/3 = 1/6$	(6.2)

The smallest displayed margin is $1/120$. In contrast, two pair activities have size $A_2^2 \ll N^{-4/3+\eta}$ and are retained at the target scale. They have no strict power saving to spend. For a prescribed final accuracy $\varepsilon_0 > 0$, one may choose $0 < \tau < \min(\varepsilon_0/4, 1/960)$ and invoke completed auxiliary estimates with total loss at most τ each. Their own input margins are chosen smaller as needed; in a product, divide the allotted loss between the factors. Thus the same symbol ε in successive statements never requires the same numerical value at every stage. Rank thresholds and fixed geometric degrees have separate roles. The fixed connected-activity estimates in reconstruction are used only through order 256; larger cardinalities have the uniform certificate estimate and a factorial boundary bound.

6 Why the quadratic approximation stops at the fourth-order scale

The quadratic reference retains the second factorial cumulant linearly. Two independent pair contributions therefore leave a fourth-order term of size α_N^2. To turn this observation into a lower bound, one must check that incompatible arithmetic configurations cannot cancel that term. The half-exponent count supplies this missing sign information. The formal term $\alpha_N^2 \Pi_{\mu_N} * \Delta^{*4}/8$ is classical in Poisson–Charlier expansions [5, 24]. The issue here is its arithmetic size, the sign of the other fourth-order contributions, and the passage from an unbounded polynomial test to the full norm. Throughout this section $\mu = \mu_N$, $\alpha = \alpha_N$, and all marginals and centered coefficients are those of the full admissible set.

6.1 Cancellation of the disconnected pair terms

Write

$$C_2^{\mathrm{s}} = \sum_{|e|=2} c(e), \qquad K_4 = \sum_{\substack{|I|=4 \\ I \text{ connected}}} c(I), \qquad K_4^+ = \sum_{\substack{|I|=4 \\ I \text{ connected, } q(I)>0}} c(I).$$

Compatibility makes every character on the relation space positive, so $K_4^+ \geq 0$.

Lemma 6.1 (The fourth factorial cumulant)**.** *For every fixed critical band and every $\varepsilon > 0$,*

$$\kappa_4 = 24K_4 + O_{C,\varepsilon}(N^{-5/3+\varepsilon}), \qquad K_4 = K_4^+ + O_{C,\varepsilon}(N^{-3/2+\varepsilon}). \tag{6.1}$$

Consequently

$$\kappa_4 + 3\alpha^2 = 3\alpha^2 + 24K_4^+ + O_{C,\varepsilon}(N^{-3/2+\varepsilon}). \tag{6.2}$$

Proof. Expanding the first four factorial moments with the individual p_i gives the exact identity

$$\begin{aligned}\kappa_4 = 24 \sum_{|I|=4} c(I) - 12(C_2^{\mathrm{s}})^2 - 24 \sum_{|T|=3} \Big(\sum_{i\in T} p_i\Big) c(T) \\ + 24 \sum_{e=\{i,j\}} (p_i^2 + p_i p_j + p_j^2) c(e) - 6 \sum_i p_i^4.\end{aligned} \tag{6.3}$$

For example, first write

$$\mathbb{E}(1+t)^Z = \prod_i (1+p_i t) \left[1 + \sum_{2 \leq |I| \leq 4} \frac{c(I) t^{|I|}}{\prod_{i\in I}(1+p_i t)} + O(t^5) \right],$$

then take 24 times the coefficient of t^4 in its formal logarithm. This is a finite coefficient calculation; no analytic logarithm of the true generating function is required.

Canonical direct decomposition gives $\sum_{|I|=4} c(I) = K_4 + D_{22}$, where D_{22} sums $c(e)c(f)$ over unordered disjoint pairs of edges whose spaces are direct. In $(C_2^{\mathrm{s}})^2/2 - D_{22}$ only repeated edges, edges sharing a vertex, and disjoint edges with a nonzero intersection remain. The positive propagation estimate, together with the product of the marginals, implies

$$\sum_{y \neq x} |c(x,y)| \ll_{C,\eta} p_x N^{\eta}.$$

Repeated edges and shared vertices consequently have total absolute weight $O(p_* A_2 N^{\eta}) = O(N^{-5/3+2\eta})$.

Apply the pair partition of Lemma T9.4 with a small fixed parameter η. A product involving a weak or exceptional edge has total weight at most $A_2(A_{\mathrm{w}} + A_{\mathrm{s}}) \ll N^{-5/3+O(\eta)}$. For two good edges from the remaining finite family of channels, fix the first edge and the offset pattern of the second compression. There are $NB^{O(1)}$ choices for the first edge. A nonzero intersection imposes an exceptional target of fixed width, giving $N^{1/2+o(1)}$ choices for the second compressed center by the elementary exceptional-target count in Proposition T1.15. The proof of that count uses only a fixed width and applies to this compression as well. Each product has weight at most $N^{-10/3+o(1)}$. Hence this last cost is $N^{-11/6+o(1)}$, and

$$D_{22} = (C_2^{\mathrm{s}})^2/2 + O(N^{-5/3+\varepsilon}).$$

The remaining terms of (6.3) are bounded by $O(p_* A_3 + p_*^2 A_2 + p_*^3 \mu)$. The triple estimate in Corollary 3.4 gives $A_3 \ll N^{-41/30+\eta}$, so each is smaller than the stated error. This proves the first identity.

The incompatible part of Proposition T12.9 at the fixed order $r = 4$ gives

$$|K_4 - K_4^{+}| \leq \sum_{\substack{|I|=4,\ I \text{ connected} \\ q(I)=0}} |c(I)| \ll_{C,\varepsilon} N^{-3/2+\varepsilon}.$$

Compatible coefficients are nonnegative. This proves the second identity and then (6.2). □

6.2 A Charlier polynomial detects the remaining mass

Theorem 6.2 (Optimal power for the exact quadratic reference)**.** *For every $C > 0$ and $\varepsilon > 0$, there are $c_{C,\varepsilon} > 0$ and $N_0(C,\varepsilon)$ such that*

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \geq c_{C,\varepsilon} N^{-4/3-\varepsilon} \tag{6.4}$$

whenever $N \geq N_0$ and $|L - \log_2 N| \leq C$. Together with Theorem 5.3 and the triple activity bound, this identifies the optimal power as $4/3$. It does not assert an asymptotic constant or a logarithmic factor for the remainder.

Proof. The arithmetic coefficient estimate gives $\alpha \asymp_C LN^{-2/3} > 0$ uniformly in the critical band. The sign argument has the margin $3/2 - 4/3 = 1/6$. Choose a fixed total error loss $\tau < 1/6$ in (6.2); then

$$\frac{N^{-3/2+\tau}}{\alpha^2} \ll_C L^{-2} N^{-1/6+\tau} \longrightarrow 0.$$

Positivity therefore yields

$$\kappa_4 + 3\alpha^2 \geq (3 - o_C(1))\alpha^2.$$

The fourth Charlier polynomial is

$$H_4(z;\mu) = (z)_4 - 4\mu(z)_3 + 6\mu^2(z)_2 - 4\mu^3 z + \mu^4 = 4![t^4]e^{-\mu t}(1+t)^z.$$

Its integral against $Q_{\mu,\alpha}$ vanishes, because the factorial generating function of that reference is $e^{\mu t}(1+\alpha t^2/2)$. On the true law its expectation equals $\kappa_4 + 3\alpha^2$. Thus, with $R = \mathcal{L}(Z_N) - Q_{\mu,\alpha}$,

$$\left| \int H_4 \, dR \right| \gg_C L^2 N^{-4/3}.$$

The test is unbounded, so a moment argument is needed before it gives a norm estimate. For every fixed integer $k > 1$, the fixed moments from Theorem T4.4 give $\int |H_4(z;\mu)|^k \, d\mathcal{L}(Z_N)(z) \ll_{C,k} 1$. The same bound holds for $|Q_{\mu,\alpha}|$, since μ stays in a compact subset of $(0,\infty)$, $\alpha = O_C(1)$, and

$$|Q_{\mu,\alpha}| \le \Pi_\mu + \frac{|\alpha|}{2} \Pi_\mu * (\delta_2 + 2\delta_1 + \delta_0).$$

Consequently Hölder's inequality for the positive measure $|R|$ gives

$$\left| \int H_4 \, dR \right| \ll_{C,k} \|R\|_1^{1-1/k}, \qquad \|R\|_1 \gg_{C,k} (L^2 N^{-4/3})^{k/(k-1)}.$$

After ε has been prescribed, choose the integer k so that $4/[3(k-1)] \le \varepsilon$. It remains fixed as N tends to infinity, and proves (6.4). The upper bound follows from the reconstruction and $A_3 \ll N^{-41/30+\eta}$, since $41/30 > 4/3$. □

The obstruction is specific to the quadratic reference. The Hermite law with the same first two factorial cumulants already contains the term $\alpha^2 \Pi_\mu * \Delta^{*4}/8$. The fourth Charlier test against that law sees κ_4 instead of $\kappa_4 + 3\alpha^2$, so the lower bound above does not establish an optimal Hermite approximation rate.

7 Completing the approximation and its consequences

The reconstruction is carried out in the algebra of signed measures. For the parameters of the model its final reference is a probability. The following positivity and continuity facts are elementary and do not depend on the law approximation. The deduction of Corollary 1.2 from the main theorem is given below.

7.1 Continuity of the reference

For nonnegative μ, μ',

$$\|\Pi_\mu - \Pi_{\mu'}\|_1 \le 2|\mu - \mu'|. \tag{7.1}$$

For example, if $\mu' = \mu + h$, couple by adding an independent Poisson variable of mean h. Its probability of being nonzero is $1 - e^{-h} \le h$, proving the inequality. It follows that

$$\|Q_{\mu,\alpha} - Q_{\mu',\alpha'}\|_1 \le 2(1 + 2|\alpha|)|\mu - \mu'| + 2|\alpha - \alpha'|. \tag{7.2}$$

Here we have expanded the difference with coefficient α on $\Pi_\mu - \Pi_{\mu'}$ and used $\|\Pi_{\mu'} * \Delta^{*2}\|_1 \le 4$.

7.2 Positivity and the coefficient of total variation

The signed representation of the reference is useful in the proof, but for the parameters of the theorem it is eventually a probability. The following exact criterion also specifies its domain for general parameters.

Lemma 7.1 (Positivity of the quadratic Poisson correction)**.** *Let $\mu > 0$, $\alpha \in \mathbb{R}$, and $\theta = \mu - \lfloor \mu \rfloor$. Then $Q_{\mu,\alpha}$ is a probability measure if and only if*

$$0 \le \alpha \le \frac{2\mu^2}{\mu + \theta(1-\theta)}. \tag{7.3}$$

Its masses are

$$Q_{\mu,\alpha}\{k\} = e^{-\mu}\frac{\mu^k}{k!}\left[1 + \frac{\alpha}{2\mu^2}\big((k-\mu)^2 - k\big)\right], \qquad k \ge 0. \tag{7.4}$$

Proof. Formula (7.4) follows by expressing the Poisson masses at $k-1$ and $k-2$ relative to the mass at k; negative-index masses are taken to be zero. The total mass is one because Δ^{*2} has total mass zero.

If $\alpha < 0$, the bracket is negative for all sufficiently large k, so positivity is impossible. For $\alpha \ge 0$, minimize the quadratic $f(k) = (k-\mu)^2 - k$ over nonnegative integers. Its successive difference is $f(k+1) - f(k) = 2(k-\mu)$. If μ is not an integer, the minimum is attained at $k = \lceil \mu \rceil$; if μ is an integer, it is attained at both $k = \mu$ and $k = \mu + 1$. In either case its value is $-\mu - \theta(1-\theta)$. Nonnegativity of the bracket is therefore exactly (7.3). □

In the theorem, $\alpha_N > 0$ for large N and $\alpha_N \to 0$, while μ_N and λ_N stay in a compact subinterval of $(0, \infty)$. Both corrected references are therefore probabilities for all sufficiently large N. The norm estimate simultaneously controls all their masses; it is not only a pointwise assertion at each fixed count.

Define, for $\lambda > 0$,

$$\mathcal{C}(\lambda) = \frac{1}{4}\|\Pi_\lambda * \Delta^{*2}\|_1 = \frac{1}{4}\mathbb{E}\left|1 - \frac{2P_\lambda}{\lambda} + \frac{P_\lambda(P_\lambda - 1)}{\lambda^2}\right|, \qquad P_\lambda \sim \mathrm{Pois}(\lambda). \tag{7.5}$$

The coefficient is explicit even when λ_N does not converge. A summable signed sequence of total mass zero has ℓ^1 norm at least twice any positive coefficient. The mass $e^{-\lambda}$ at zero of $\Pi_\lambda * \Delta^{*2}$ and its convolution norm therefore give

$$\frac{e^{-\lambda}}{2} \le \mathcal{C}(\lambda) \le 1. \tag{7.6}$$

7.3 Completion of the main theorem

The arithmetic and probabilistic estimates now meet at a single quantity, the absolute third-order activity. We complete the main theorem, then record what its precision allows for unbounded scalar observables and for the choice of mean.

Proof of Theorem 1.1. Write $\varepsilon_{\mathrm{out}} = \varepsilon$ for the requested final margin and choose $0 < \tau < \min(\varepsilon_{\mathrm{out}}/4, 1/960)$. Invoke Theorem 5.3 with accuracy τ and Corollary 3.4 with accuracy τ. The triple contribution is

$$N^{\tau} A_3 \ll N^{-41/30+2\tau} = N^{-4/3-(1/30-2\tau)}.$$

Since $1/30 - 2\tau > 0$ and the other reconstruction term is $N^{-4/3+\tau}$, this proves the upper bound. Theorem 6.2 gives the lower bound for the same exact parameters, using its own fixed Charlier interpolation order.

The mean estimate is Proposition T2.1. The pair asymptotic (1.5) shows that $\alpha_N > 0$, $\alpha_N \to 0$, and μ_N stays in a compact subset of $(0, \infty)$. Lemma 7.1 therefore makes the reference a probability measure for sufficiently large N. Every auxiliary accuracy parameter and fixed order was chosen after C, ε and before N. The conclusion is uniform on the entire fixed critical band. □

7.4 Polynomial weights and fixed moments

Corollary 7.2 (Every fixed polynomial weight)**.** *For every fixed integer $m \geq 0$ and $\varepsilon > 0$,*

$$\sum_{k\geq 0}(1+k)^m \left|\mathbb{P}(Z_N = k) - Q_{\mu_N,\alpha_N}\{k\}\right| \ll_{C,m,\varepsilon} N^{-4/3+\varepsilon}. \tag{7.7}$$

For every fixed $r \geq 2$, consequently,

$$\mathbb{E}(Z_N)_r = \mu_N^r + \binom{r}{2}\mu_N^{r-2}\alpha_N + O_{C,r,\varepsilon}(N^{-4/3+\varepsilon}). \tag{7.8}$$

The equalities for $r = 0, 1, 2$ hold without a remainder when the right side is interpreted in its natural form.

Proof. Let $R = \mathcal{L}Z_N - Q_{\mu_N,\alpha_N}$. For every fixed integer $v > 1$, Hölder gives

$$\sum_k (1+k)^m |R\{k\}| \leq \|R\|_1^{1-1/v} \left(\sum_k (1+k)^{mv}|R\{k\}|\right)^{1/v}.$$

The second factor is uniformly bounded. For the true law this follows from the independent fixed-moment proof in Theorem T4.4; for the reference it follows by dominating its variation by a fixed sum of shifted Poisson laws, as in Theorem 6.2. Choose v sufficiently large after ε and then a sufficiently small margin in the main theorem. This proves (7.7). The factorial generating function of $Q_{\mu,\alpha}$ is $e^{\mu t}(1 + \alpha t^2/2)$; its coefficient of t^r, multiplied by $r!$, gives the moment formula. □

This interpolation concerns the scalar count and fixed polynomial degrees.

Corollary 7.3 (No retained start)**.** *Uniformly in a fixed critical band,*

$$\mathbb{P}(Z_N = 0) = e^{-\mu_N}\left(1 + \frac{\alpha_N}{2}\right) + O_{C,\varepsilon}(N^{-4/3+\varepsilon}).$$

Proof. Evaluate the approximation at zero, using $(\Pi_\mu * \Delta^{*2})\{0\} = e^{-\mu}$. □

The event is the absence of a retained start in I_N. It does not exclude a long run that started before N.

7.5 Why the mean must be specified at this precision

Proposition 7.4 (The quantized mean bias)**.** *Let η_x indicate one-window compatibility and put $h_x = L - \dim S_x$. Then*

$$\mu_N - \lambda_N = pM_{N,L}, \qquad M_{N,L} = \sum_{x\in I_N}(\eta_x 2^{h_x} - 1) \in \mathbb{Z}. \tag{7.9}$$

Thus either $\mu_N = \lambda_N$ or $|\mu_N - \lambda_N| \geq p \asymp_C N^{-1}$. Keeping the same exact α_N, one has

$$\|\mathcal{L}Z_N - Q_{\lambda_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{-1+\varepsilon}.$$

On every sufficiently large admissible pair (N, L) for which $M_{N,L} \neq 0$,

$$\|\mathcal{L}Z_N - Q_{\lambda_N,\alpha_N}\|_1 \asymp_C |\mu_N - \lambda_N|. \tag{7.10}$$

No assertion that $M_{N,L} \neq 0$ at infinitely many scales is included.

Proof. Sum the one-window formula $\mathbb{E}J_x = \eta_x p 2^{h_x}$ over the exactly N centres to obtain (7.9). For two means in the compact range in question, (7.2) gives

$$\|Q_{\mu,\alpha_N} - Q_{\lambda,\alpha_N}\|_1 \ll_C |\mu - \lambda|.$$

The reverse bound follows from the mass at zero, $e^{-\theta}(1+\alpha_N/2)$, by the mean value theorem. The upper bound for the nominal reference follows from the main theorem and $|\mu_N - \lambda_N| \ll N^{-1+\varepsilon}$.

If $M_{N,L} \neq 0$, the mean difference is at least p. Choose a fixed margin less than $1/3$ in the main theorem; its remainder is then $o(p)$. The triangle and reverse triangle inequalities transfer the two-sided reference comparison to (7.10). □

Irregular compatible and incompatible windows contribute with opposite signs to $M_{N,L}$. Their presence alone does not prove a nonzero total bias.

7.6 The total-variation coefficient

Proof of Corollary 1.2. Theorem 1.1 and the reverse triangle inequality first give

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_{\mu_N}) = \alpha_N \mathcal{C}(\mu_N) + O_{C,\varepsilon}(N^{-4/3+\varepsilon}).$$

Use $|\mu_N - \lambda_N| \ll N^{-1+\varepsilon}$ and (7.2) to obtain

$$\mathcal{L}Z_N - \Pi_{\lambda_N} = \frac{\alpha_N}{2}\Pi_{\lambda_N} * \Delta^{*2} + O_{\ell^1,C,\varepsilon}(N^{-1+\varepsilon}).$$

The reverse triangle inequality, followed by the factor $1/2$ converting the ℓ^1 norm to total variation, yields

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_{\lambda_N}) = \alpha_N \mathcal{C}(\lambda_N) + O_{C,\varepsilon}(N^{-1+\varepsilon}). \tag{7.11}$$

The pair asymptotic from Theorem 3.1 now gives

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_{\lambda_N}) \sim \mathcal{C}(\lambda_N)\frac{r_N+2}{18\,2^{r_N/3}}\,\lambda_N^{5/3} L N^{-2/3}, \qquad r_N = L \bmod 3. \tag{7.12}$$

The lower bound in (7.6) is uniform on the fixed critical band, and the remainder is $o(LN^{-2/3})$. This proves the sharp order asserted in the corollary. The equivalence is relative to the displayed oscillating expression; no convergence of its phase or intensity factors is required. □

Replacing α_N by its leading equivalent proves (7.12). It does not by itself preserve the error $O(N^{-1+\varepsilon})$ in (7.11), which uses the exact second factorial cumulant.

Proposition T14.1 gives $\|\Pi_1 * \Delta^{*2}\|_1 = 3/e$, hence $\mathcal{C}(1) = 3/(4e)$. Along $N = 2^L$ with $3 \mid L$, the pair coefficient therefore gives

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_1) \sim \frac{1}{12e}\,L N^{-2/3}. \tag{7.13}$$

This is the coefficient in the example announced in the introduction.

The total-variation asymptotic uses the exact-marginal reconstruction and the explicit pair coefficient. The independent positivity and reference-continuity estimates do not use the reconstruction.

8 Explicit references and the first omitted channel

The exact second factorial cumulant retains every pair interaction. An explicit reference can instead retain the complete contributions of all rational channels up to a chosen height. The first omitted height then determines its error, including the leading coefficient. This gives a different optimal exponent for each finite truncation, even though all these references approximate the same random variable.

For coprime integers $b/2 < a < b$, $b \geq 3$, retain the finite sums $S_{a,b}(L)$ from (3.2) and put

$$w_{a,b} = \frac{2(2a-b)}{ab}, \qquad T_b(L) = \sum_{\substack{b/2<a<b \\ (a,b)=1}} w_{a,b} S_{a,b}(L), \tag{8.1}$$
$$\widehat{\alpha}_N^{(\leq h)} = \frac{\lambda_N^2}{N} \sum_{b=3}^{h} T_b(L), \qquad h \geq 3.$$

The factor two in $w_{a,b}$ counts the two orientations of a pair. Here and throughout this section, the integer h is fixed before N. In particular, $\widehat{\alpha}_N^{(\leq 4)} = \widehat{\alpha}_N$ from (3.10).

Theorem 8.1 (The optimal error of every fixed height truncation)**.** *Fix $h \geq 3$ and $C > 0$, and write $b = h+1$ and $r = L \bmod b$. For every $\varepsilon > 0$, uniformly in the critical band,*

$$\alpha_N - \widehat{\alpha}_N^{(\leq h)} = \frac{\lambda_N^2}{N} T_b(L) + O_{h,C,\varepsilon}(N^{-b/(b+1)+\varepsilon}), \tag{8.2}$$
$$\mathcal{L}(Z_N) - Q_{\lambda_N, \widehat{\alpha}_N^{(\leq h)}} = \frac{\lambda_N^2 T_b(L)}{2N} \Pi_{\lambda_N} * \Delta^{*2} + O_{\ell^1,h,C,\varepsilon}(N^{-b/(b+1)+\varepsilon}). \tag{8.3}$$

Define the positive number

$$c_b(r) = \frac{b+r-1}{b^2\, 2^{r/b}} \sum_{\substack{b/2<a<b \\ (a,b)=1}} \frac{(2a-b)(b-a)}{a}. \tag{8.4}$$

The reference is a probability measure for all sufficiently large N, and

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), Q_{\lambda_N, \widehat{\alpha}_N^{(\leq h)}}) \sim \frac{c_b(r)}{4} \lambda_N^{2-1/b} \|\Pi_{\lambda_N} * \Delta^{*2}\|_1 \, L N^{-1+1/b}. \tag{8.5}$$

The ratio tends to one uniformly over the critical band with the displayed moving parameters retained. Thus the optimal power exponent of this specified reference is $h/(h+1)$.

Before proving the theorem, we account for the nonaligned pairs. The unweighted count alone would not suffice: a pair carrying many relations has a correspondingly larger joint probability.

Lemma 8.2 (Weighted nonaligned pairs)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{x,y \in G_N,\ |x-y|>L \\ (x,y)\ \text{nonaligned}}} (2^{\rho(x,y)} - 1) \ll_{C,\varepsilon} N^{1+\varepsilon}. \tag{8.6}$$

Consequently the sum of the absolute covariances on these pairs is $O_{C,\varepsilon}(N^{-1+\varepsilon})$.

Proof. Use the large-prime component decomposition (T2.5), so that $\rho(x,y) \leq D(x,y) + c(x,y)$ and $D(x,y) = o(B)$ uniformly. Fix $\eta > 0$ temporarily. On $c(x,y) < \eta B$, the weight is at most $2^{\eta B + o(B)}$. A pair in G_N with $\rho(x,y) > 0$ has a nonzero intersection between its two effective spaces. The pair count of Corollary T6.3 therefore bounds the contribution of this sector by $N^{1+\eta+\varepsilon'+o(1)}$. On $c(x,y) \geq \eta B$, Lemma T1.8 already bounds the entire weighted sum by $N^{1+\varepsilon'}$. Choose η and then ε' inside the prescribed margin.

On G_N both marginals are p. The affine identity (T2.1) gives $|\operatorname{Cov}(J_x, J_y)| \leq p^2(2^{\rho(x,y)} - 1)$, including incompatible pairs. Since $p \asymp_C N^{-1}$, this proves the last assertion. □

Proof of Theorem 8.1. We first prove, for every fixed integer $m \geq 3$,

$$\alpha_N - \widehat{\alpha}_N^{(\leq m)} = O_{m,C,\varepsilon}(N^{-m/(m+1)+\varepsilon}). \tag{8.7}$$

The regular-set transfer in Proposition T2.1, justified by the independent fixed-moment bound of Theorem T4.4, changes the second factorial cumulant by $O(N^{-1+\varepsilon})$. On G_N, that cumulant is the sum of covariances over ordered distinct pairs minus $|G_N|p^2$. The last term is $O(N^{-1})$.

Strict contacts are incompatible and number $O(NB)$, so their covariance mass is $O(B/N)$. At separation exactly L, the union of the two windows lies in an interval of length $O(B)$ and has formal rank $2L - o(L)$, by Lemma T1.3; this band has the same $N^{-1+o(1)}$ cost. Lemma 8.2 treats the separated nonaligned pairs.

For aligned separated pairs, reduced slopes are unique for sufficiently large N: their separation is at least B^{-6}, whereas the allowed offset changes the ratio of the starts by only $B^{O(1)}/N$. There are $NB^{O(1)}$ aligned hosts altogether. A slope of height one has no exact unit for separated windows and hence only residual nullity $o(L)$ by Lemma T1.9. The slope $1:2$ is confined to $B^{O(1)}$ pairs at the dyadic endpoints; its total mass is at most $N^{-3/2+o(1)}$. Every remaining reduced slope is an interior channel $b/2 < a < b$, $b \geq 3$.

For each fixed channel of height at most m, normalize to $bV_x \cup aV_y$. Lemma T3.5 gives its ordered contribution $p^2 N w_{a,b} S_{a,b}(L) + O_{m,C,\varepsilon}(N^{-1+\varepsilon})$. Changing from the good-pair joint mass to the covariance costs at most $O_m(p^2 NB)$. The omitted pairs of bad normalization have total weight $N^{-3/2+1/b+o(1)}$, as in that lemma's proof, and are smaller still. Only finitely many channels have been retained.

For all remaining heights $b \geq m+1$, including heights varying up to B^3, the uniform aligned rank bound gives $\rho(x,y) \leq L/(m+1) + o(L)$. Their total contribution, in absolute value, is at most

$$p^2 NB^{O(1)} 2^{L/(m+1)+o(L)} = N^{-m/(m+1)+o(1)}.$$

This proves (8.7). Apply it at $m = h+1$ and separate the last retained term to obtain (8.2).

The mean estimate of Proposition T2.1 gives $|\mu_N - \lambda_N| \ll N^{-1+\varepsilon'}$. Coupling Poisson variables shows $\|\Pi_u - \Pi_v\|_1 \leq 2|u-v|$. Since $\alpha_N = o(1)$,

$$\|Q_{\mu_N,\alpha_N} - Q_{\lambda_N,\alpha_N}\|_1 \ll |\mu_N - \lambda_N|.$$

Theorem 1.1, this estimate, and the exact linearity of $Q_{\lambda,a}$ in a give (8.3). Both the mean error of power one and the exact-reference error of power $4/3$ are smaller than $N^{-b/(b+1)+\varepsilon}$ for this fixed b.

To identify the leading term, write $L = bq + r$, $0 \leq r < b$. The finite identity (3.2), with $s = -1$, yields

$$S_{a,b}(L) = 2^{q-1}\big[(b-a)(b+r-1)q + O_b(1)\big] + O_b(L).$$

As $q \sim L/b$ and $2^q = 2^{-r/b}(N/\lambda_N)^{1/b}$, summation gives

$$\frac{\lambda_N^2}{N} T_b(L) \sim c_b(r) \lambda_N^{2-1/b} L N^{-1+1/b}. \tag{8.8}$$

Every summand in (8.4) is positive, and $a = b-1$ is always present. The phase ranges over a fixed finite set and λ_N over a compact subset of $(0,\infty)$. Taking $\varepsilon < 1/(b(b+1))$ makes the error in (8.3) negligible relative to its leading term. The reverse triangle inequality then gives the norm equivalent; division by two gives (8.5). The factor $1/4$ combines this division with the quadratic coefficient $1/2$.

Finally, $\widehat{\alpha}_N^{(\leq h)} \sim \alpha_N > 0$ and tends to zero. The positivity criterion of Lemma 7.1 therefore applies uniformly on the compact range of λ_N. The leading coefficient in (8.5) is bounded above and away from zero, since $\|\Pi_{\lambda_N} * \Delta^{*2}\|_1 \geq e^{-\lambda_N}$. This proves optimality of the stated power for the specified reference. □

8.1 What the first omitted height contributes

The four channels of heights at most five are $2:3$, $3:4$, $3:5$, and $4:5$. Thus the next explicit correction after (3.10) has a short closed formula.

Corollary 8.3 (The error of the two-channel reference)**.** *Write $L = 5q + r$, $0 \le r < 5$, and, for $q \ge 1$, set*

$$\begin{aligned} M_5(N,L) &= \frac{\lambda_N^2}{N}\left\{\frac{2}{15}S_{3,5}(L) + \frac{3}{10}S_{4,5}(L)\right\} \\ &= \frac{\lambda_N^2}{30N}\left\{2^{q-1}\big[17(r+4)q + 13r^2 + 87r + 620\big] - 113L - 127\right\}. \end{aligned} \tag{8.9}$$

Then

$$\mathcal{L}(Z_N) - Q_{\lambda_N,\widehat{\alpha}_N} = \frac{M_5(N,L)}{2}\Pi_{\lambda_N} * \Delta^{*2} + O_{\ell^1,C,\varepsilon}(N^{-5/6+\varepsilon}), \tag{8.10}$$

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), Q_{\lambda_N,\widehat{\alpha}_N}) \sim \frac{17(r+4)}{1200\,2^{r/5}}\lambda_N^{9/5}\|\Pi_{\lambda_N} * \Delta^{*2}\|_1\, LN^{-4/5}. \tag{8.11}$$

The four-channel coefficient is $\widehat{\alpha}_N^{(\le 5)} = \widehat{\alpha}_N + M_5(N,L)$. Its reference has optimal exponent $5/6$, with leading phase constant $c_6(r) = (r+5)/(45\,2^{r/6})$, where now $r = L \bmod 6$.

Proof. The finite identity gives

$$\begin{aligned} S_{3,5}(L) &= 2^{q-1}\big[2(r+4)q + r^2 + 6r + 38\big] - 8L - 7, \\ S_{4,5}(L) &= 2^{q-1}\big[(r+4)q + r^2 + 7r + 52\big] - 9L - 11. \end{aligned}$$

Their weighted sum proves (8.9). Theorem 8.1 at $h = 4$ gives the first two conclusions. At height six the only reduced interior channel is $5:6$; substitution in (8.4) gives c_6 and the assertion for the four-channel reference. □

For example, if $N = 2^L$ and $5 \mid L$, the error of the two-channel reference is asymptotic in total variation to $17LN^{-4/5}/(100e)$, because $\|\Pi_1 * \Delta^{*2}\|_1 = 3/e$. The table records how successive complete channels change the error.

Height h	Retained channels	Optimal power	First omitted channels
3	$2:3$	$3/4$	$3:4$
4	$2:3,\ 3:4$	$4/5$	$3:5,\ 4:5$
5	$2:3,\ 3:4,\ 3:5,\ 4:5$	$5/6$	$5:6$
h fixed	all reduced interior heights $\le h$	$h/(h+1)$	height $h+1$

The full coefficient in (8.1) has a phase dividing $\mathrm{lcm}(3,\dots,h)$, whereas its leading omitted term uses only $L \bmod (h+1)$, together with λ_N. The theorem permits any fixed height, with constants depending on that height. It makes no assertion for a height growing with N. In particular it does not identify an obstruction of power one for other explicit references: the stated optimality concerns precisely the finite truncations defined in (8.1).

8.2 Positive Hermite references

Corollary 8.4 (Hermite approximation with single and double jumps)**.** *For all sufficiently large N, let U_N, V_N be independent Poisson variables with means $\mu_N - \alpha_N$ and $\alpha_N/2$, respectively. Then*

$$d_{\mathrm{TV}}(\mathcal{L}(Z_N), \mathcal{L}(U_N + 2V_N)) \ll_{C,\varepsilon} N^{-4/3+\varepsilon}.$$

For every fixed $h \geq 3$, the entirely explicit reference

$$\widehat{U}_N + 2\widehat{V}_N, \qquad \widehat{U}_N \sim \mathrm{Pois}(\lambda_N - \widehat{\alpha}_N^{(\leq h)}), \qquad \widehat{V}_N \sim \mathrm{Pois}(\widehat{\alpha}_N^{(\leq h)}/2),$$

with independent summands has optimal power $h/(h+1)$ and the same leading distance equivalent as the corresponding quadratic reference in Theorem 8.1. In particular the two-channel choice $h = 4$ has error $O(N^{-4/5+\varepsilon})$. No matching lower bound is asserted here for the exact-parameter Hermite reference.

Proof. Both means are nonnegative for large N. In the convolution algebra, the law of $U_N + 2V_N$ is

$$\exp_*\left(\mu_N\Delta + \frac{\alpha_N}{2}\Delta^{*2}\right).$$

Expanding the second exponential to first order gives Q_{μ_N,α_N} with an ℓ^1 error $O_C(\alpha_N^2)$. Indeed $\|\Delta^{*2}\|_1 = 4$, and the exponential remainder is bounded by $\frac{1}{2}(2\alpha_N)^2 e^{2\alpha_N}$. Since $\alpha_N^2 = N^{-4/3+o(1)}$, the main theorem proves the claim. For each fixed-height explicit reference, the coefficient is positive and tends to zero uniformly in the critical band, while λ_N stays bounded away from zero. Thus both proposed Poisson means are nonnegative for all sufficiently large N. Repeat the same expansion with parameters $(\lambda_N, \widehat{\alpha}_N^{(\leq h)})$, and apply Theorem 8.1. The additional norm error is $O_C((\widehat{\alpha}_N^{(\leq h)})^2) = N^{-4/3+o(1)}$, smaller than the positive first omitted channel term of order $LN^{-h/(h+1)}$. The reverse triangle inequality preserves its equivalent and hence its optimal exponent. □

The double jump represents the leading pair interaction at the level of the scalar law. The contribution here is the precision of the Hermite approximation for Z_N. Its double jump intensity uses either the exact second factorial cumulant or a complete finite channel sum. This is a representation of the reference law; it does not decompose the actual run starts into independent components. Keeping only the leading equivalent of that coefficient does not preserve the precision of a fixed-height truncation. The exact Hermite law already contains the quartic term $\alpha_N^2\Pi_{\mu_N} * \Delta^{*4}/8$, so the lower bound for the quadratic reference does not transfer to it.

This distinction is already present in the locally dependent setting of Čekanavičius and Vellaisamy [24, Theorem 3.1, (3.5), and Theorem 3.2, (3.6)]. Their quadratic reference is $\Pi_{\Gamma_1} * (\delta_0 + \Gamma_2\Delta^{*2})$, whereas the exponential reference is $\exp_*(\Gamma_1\Delta + \Gamma_2\Delta^{*2})$, with $\Gamma_2 = (\mathrm{Var}\, Z - \mathbb{E}Z)/2$. The quadratic estimate contains an additional squared error term that is absent from the exponential estimate. These results concern sums reduced to one-dependence under specified moment conditions. We use this comparison to place the references in context; the arithmetic estimates here provide their approximation to the multiplicative run count.

9 What the proof makes reusable

The exponent 4/3 results from the relation between two scales: the leading pair mass is $N^{-2/3+o(1)}$, and the quadratic reference omits a contribution formed from two such interactions. The arithmetic work shows that all other terms relevant to the reconstruction can be controlled at this precision. The proof also separates several choices that can be made in other rare-event problems.

9.1 Choose a reference before estimating the error

The independent Bernoulli reference retains every exact marginal. Its centered expansion has no one-site term, and repeated indices are already accounted for. The retained pair correction then removes the complete second-order contribution. This makes the quantity to be bounded a residual interaction, with a specified sign and normalization.

A homogeneous envelope provides a second useful reduction. The joint probability of the homogeneous constraints bounds the actual joint probability, the exact marginal rank weight and the absolute centered activity. Geometric counting can therefore be performed once and used for each of these weights. At a fixed arity r, the comparison with the nominal weight costs at most 2^r. That fixed factor can be absorbed into the constant of a fixed-order estimate. The estimates that sum over unbounded arities keep their own weights and quantifiers.

9.2 Stop where the remaining dependence is still positive

The specialization of the abstract lace expansion [21] in Section 4 retains a hereditary class of subsets with computable joint probabilities. At the first excluded prefix, all later variables remain in a positive conditional measure. Its total mass is the joint probability of the prefix. The boundary therefore has a direct probabilistic weight even though the retained expansion is signed. Incompatible pair and triple terms must still be combined before absolute values are taken.

To use this identity in another model, one needs a retained class that matches the chosen reference and quantitative control of the first excluded configurations. Here compression makes a dependent pair one affine block. Conditional arithmetic counts control a fixed number of blocks, and congruence certificates control the large-cardinality part. The reconstruction estimate records exactly what these inputs accomplish:

$$\|\mathcal{L}(Z_N) - Q_{\mu_N,\alpha_N}\|_1 \ll N^{\varepsilon} A_3 + N^{-4/3+\varepsilon}.$$

The finite identity itself does not give the arithmetic bounds needed on its right-hand side.

9.3 Separate fixed orders from a uniform tail

A fixed-order theorem allows its constant and scale threshold to depend on the order. It cannot be summed up to an order growing with N without further information. The proof instead chooses a finite cutoff after the desired accuracy, treats the finitely many lower orders by their individual estimates, and uses one uniform weighted tail above that cutoff.

Proposition T13.1 illustrates this arrangement. Finite layers give a half power for each effective block, while the independent tail gives every power strictly below 2. Their combination yields the exponent $\min(s/2, 2)$, with an arbitrary fixed margin and a common constant for all $s \geq 2$. This is a bound on a deterministic sum of independent counting weights. It is not an exponential-tail statement for the law of Z_N.

The independent fixed-moment proof has a different use: it turns a polynomial test into a norm lower bound and gives fixed polynomially weighted consequences of the main theorem. The finite-difference example in Section T15 shows why even very strong information at each fixed moment order does not by itself provide a polynomial rate for the complete law.

9.4 Test the approximation that was actually retained

The fourth Charlier polynomial detects $\kappa_{4,N} + 3\alpha_N^2$ after subtraction of the quadratic reference. The nonnegative compatible part of the quartic identity prevents cancellation at the scale of α_N^2. For the exact Hermite reference the same test instead detects $\kappa_{4,N}$, because its exponential already contains the two-pair term. A lower bound is therefore tied to a specified reference, not just to the underlying dependent count.

The full equivalent for the exact quadratic remainder, the full constant for the third factorial cumulant, and the sharp error of the exact Hermite reference remain open in this work. So do the errors after cubic or quartic correction and the optimization of total variation over all Poisson means. The explicit channel coefficients determine reference laws, but the constants in

the asymptotic remainders and the thresholds in N are not supplied numerically. Applying the method elsewhere requires counterparts of the conditional counts, compatibility cancellations and uniform boundary estimate; the stopping identity alone does not impose a universal approximation exponent.

Formal proofs and this edition

The associated development is available at https://github.com/Vulkin-prog/paper-p1-lean. The aligned edition of 12 November 2026 maps all 118 numbered results (26 in the article and 92 in the companion) to Lean proofs checked with Lean 4.36.0-rc2 and a pinned Mathlib revision. The source and compiled-type index records every clause and inherited hypothesis, including the printed conditional finite-transfer principle. Explicit interfaces cover the homogeneous envelope, the strengthened fixed-arity bounds, positive Palm costs for arbitrary target windows, and the support count uniform in its lower cardinality. The repository preserves the received edition, earlier manuscripts, proof-replay records and kernel axiom audits. Results outside the present scalar-count scope remain in the historical library. Palomar submissions remain on hold; these checks are not registry certification.

Statement on AI assistance

ChatGPT (GPT-6 Astra) was used to organize the research material, develop and rewrite mathematical arguments, inspect references, and prepare the article and its technical companion. Parallel AI-assisted reviews examined selected identities, constants, proof dependencies and the present restructuring. The associated Lean development also used AI assistance for proof construction and statement comparison. AI-assisted review and machine checking are distinct from independent human verification. The author remains responsible for the mathematical claims, their correspondence with the formal statements, and any submission.

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